Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/21/2016
PI Name:	Mason, Christopher Ph.D.		
Project Title:	The Landscape of DNA and RNA Methylation Before, During, and After Human Space Travel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chm2042@med.cornell.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-668-1448
Organization Name:	Weill Medical College of Cornell University		
PI Address 1:	1305 York Ave, Room Y13-04, Box 140		
PI Address 2:			
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/15/2014	**End Date:**	04/14/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
Contact Email:	jbcharle@gmail.com		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 4/14/2018 per NSSC information (Ed., 3/21/17)		
Key Personnel Changes/Previous PI:	August 2016: Francine Garrett-Bakelman is now a Co-Investigator.		
COI Name (Institution):	Melnick, Ari M.D. (Weill Medical College of Cornell University) Grills, George Ph.D. (Cornell University) Garrett-Bakelman, Francine M.D., Ph.D. (Weill Cornell Medicine)		
Grant/Contract No.:	NNX14AH50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Next-generation sequencing (NGS) technologies have recently been used to detect and classify epigenetic changes at an unprecedented level. Epigenetic patterns help define cellular phenotypes and can even be used to more accurately sub-classify human tumors such as acute myeloid leukemias (AML). Indeed, since some phenotypes such as cancer and aging can be better differentiated with epigenetic signatures (vs. purely genetic), epigenetic profiling is the focus of large-scale NGS efforts at the National Institutes of Health (NIH) like the Epigenomics Roadmap. There is also evidence that epigenetic changes accompany DNA damage and that sites of methylation correspond to DNA damage sites. Finally, there is an emerging field of "epitranscriptomics," where modifications of RNA, like the epigenetic modification of DNA, have been shown to impact brain development, gene regulation, and protein translation rates. Such ongoing research in DNA and RNA methylation has shown that these mediators of gene expression and modification represent critical regulatory layers in cell biology and genetics. Yet, these key epigenetic mechanisms of cellular regulation have barely, or never, been studied in the context of human space travel or during various gravity transitions ("G-transitions"). While previous work has addressed gene expression and proteomic dynamics during space travel, this Human Exploration Research Opportunities (HERO) solicitation from NASA regarding the Kelly twins provides an extraordinary opportunity to finally examine the impact of space travel on epigenetic, epitranscriptomic, and expression changes for identical twins. We propose to generate genome-wide maps of DNA and RNA methylation sites for the twins, as well a complete catalog of coding and noncoding RNA expression, to generate a 13 time-point, integrated portrait of gene regulation before, during, and after space travel. We will use established biochemical methods and open-source algorithms to profile these samples, which will reveal both the inter-twin differences and intra-person temporal changes in DNA and RNA. This thorough, genome-wide molecular portrait of the epigenetic, epitranscriptomic, and gene regulatory landscape of these twins will help address the potential epigenetic and long-term genome regulatory risk of space travel, and also discern if there is similarity of the twins' epigenetic changes to other phenotypes like aging and cancer. Specifically, we will create: (Aim 1) – Genome-wide epigenetic profiles of DNA methylation changes. We will use enhanced reduced representation bisulfite sequencing (eRRBS) to discover the precise sites of differentially methylated cytosines (DMCs) and differentially methylated regions (DMRs) for these twins. We will also estimate the number of cells (the clonality) that harbor methylation changes using our methylkit, eDMR, and MethClone algorithms. (Aim 2) – A comprehensive catalog of coding and noncoding, small and large RNAs. We will use total RNA-sequencing (RNA-seq) with ribo-depletion of the RNA samples to characterize the coding RNA and the non-coding RNAs, as well as micro-RNA (miRNA) sequencing of the small RNAs. We will then examine the impact of the DMCs and DMRs from Aim #1 on gene expression, alternative splicing of expressed genes, fusion events, and catalog the number of non-coding RNAs active in the genome during space travel. (Aim 3) – Transcriptome-wide maps of RNA methylation sites. We will use our published Methylated RNA ImmunoPrecipitation sequencing (MeRIP-seq) methods and algorithms (MeRIPPer, r-make) to examine the sites of RNA methylation that occur during space travel, and integrate the epitranscriptomic changes with those that appear in the gene expression and epigenetic levels. Having such a thorough molecular perspective will serve the long-term goals of the NASA program and also drive basic biological discoveries about the human genome's response to space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have used methclone to look at shifts in clonality of methylation and also published this work.
	Introduction: This study is part of an integrated compilation of multiple studies led by multiple PIs at multiple centers which take advantage of a very unique opportunity to look at the effects of space travel on identical twins, one of whom experiences space travel for one year; the other remains Earth-bound for that same year. This project investigates how an astronaut's landscape of DNA and RNA methylation changes in response to acute and prolonged exposure to microgravity and other spaceflight stressors and after return to Earth, whether these changes differ relative to the astronaut's twin brother, and how they relate to physiological, molecular, genetic, and biochemical changes. Next-generation sequencing (NGS) technologies have recently been used to detect and classify epigenetic changes at an unprecedented level. Epigenetic patterns help define cellular phenotypes and can even be used to more accurately sub-classify human tumors such as acute myeloid leukemias (AML). Indeed, since some phenotypes such as cancer and aging can be better differentiated with epigenetic signatures (vs. purely genetic), epigenetic profiling is the focus of large-scale NGS efforts at the NIH like the Epigenomics Roadmap. There is also evidence that epigenetic changes accompany DNA damage and that sites of methylation correspond to DNA damage sites. Finally, there is an emerging field of "epitranscriptomics," where modifications of RNA, like the epigenetic modification of DNA, have been shown to impact brain development, gene regulation, and protein translation rates. Such ongoing research in DNA and RNA methylation has shown that these mediators of gene expression and modification represent critical regulatory layers in cell biology and genetics. Yet, these key epigenetic mechanisms of cellular regulation have barely, or never, been studied in the context of human space travel or during various gravity transitions ("G-transitions"). While previous work has addressed gene expression and proteomic dynamics during space travel, this HERO solicitation from NASA regarding the Kelly twins provides an extraordinary opportunity to finally examine the impact of space travel on epigenetic, epitranscriptomic, and expression changes for identical twins. We propose to generate genome-wide maps of DNA and RNA methylation sites for the twins, as well a complete catalog of coding and noncoding RNA expression, to generate a 13 time-point, integrated portrait of gene regulation before, during, and after space travel. We will use established biochemical methods and open-source algorithms to profile these samples, which will reveal both the inter-twin differences and intra-person temporal changes in DNA and RNA. This thorough, genome-wide molecular portrait of the epigenetic, epitranscriptomic, and gene regulatory landscape of these twins will help address the potential epigenetic and long-term genome regulatory risk of space travel, and also discern if there is similarity of the twins' epigenetic changes to other phenotypes like aging and cancer. Specific Aims: Aim 1 – Genome-wide epigenetic profiles of DNA methylation changes

Task Progress:	Aim 2 – A comprehensive catalog of coding and noncoding, small and large RNAs Aim 3 – Transcriptome-wide maps of RNA methylation sites Since the initiation of the collaborative project, it was decided that aim one will be performed by Dr. Andrew Feinberg to avoid duplicated efforts. He will be performing genome-wide DNA methylation profiling. We will instead perform genome-wide hydroxymethylation profiling. These two approaches complement each other and yield additional information from the same sample collections about the epigenomic landscape. Materials and Methods: We will collaborate with Dr. Andrew Feinberg to obtain genome-wide DNA cytosine methylation data from his planned experiments. We will analyze the data to determine the precise sites of differentially methylated cytosines (DMCs) and differentially methylated regions (DMRs) for these twins. The number of cells (the clonality) that harbor methylation changes will be estimated using our methylkit, eDMR, and MethClone algorithms. We will utilize click-based chemistry to then profile genome-wide hydroxymethylation. Analysis tools which we developed for DNA methylation analysis can also be used to determine genomic regions enriched for hydroxymethylation. We will use total RNA-sequencing (RNA-seq) with ribo-depletion of the RNA samples to characterize the coding RNA and the noncoding RNAs, as well as micro-RNA (miRNA) sequencing of the small RNAs. We will then examine the impact of the DNA methylation and hydroxymethylation patterning determined in Aim #1 on gene expression, alternative splicing of expressed genes, fusion events, and catalog the number of non-coding RNAs active in the genome during space travel. Our published Methylated RNA ImmunoPrecipitation sequencing (MeRIP-seq) methods and algorithms (MeRIPPer, rmake) will inform examination of the sites of RNA methylation that occur during space travel, and integrate the epitranscriptomic changes with those that appear in the gene expression and epigenetic levels. Results: Methods for processing freshly collected pre-flight samples were determined and validated using flow cytometry for cell purity verification. Furthermore, downstream high molecular weight nucleic acid extractions for use by our group, as well as others (Drs. Michael Snyder and Emmanuel Mignot) were tested and validated for utilization in proposed studies. Pre-Flight samples on both the inflight and ground subject were collected and processed. Plasma and Cells not needed at this time for use were banked for future processing. Methods for processing of frozen vials collected on the International Space Station (ISS) are currently being validated. Discussion: Samples collected to date have all been processed and nucleic acids extracted are of high quality. All samples are amenable for use in downstream assays. Only one cell type (CD19) did not yield sufficient quantities for all proposed studies in all participating laboratories. We have distributed cells, DNA and RNA to ourselves, and Drs. Snyder and Mignot's laboratories. Currently, we are processing the RNA samples for transcriptomic profiling. Proposed protocols for data sharing are being examined with collaborating investigators. We have also completed the last of the blood collections for the twins.
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Peer-reviewed Journals	Li S, Garrett-Bakelman FE, Chung SS, Sanders MA, Hricik T, Rapaport F, Patel J, Dillon R, Vijay P, Brown AL, Perl AE, Cannon J, Bullinger L, Luger S, Becker M, Lewis ID, To LB, Delwel R, Löwenberg B, Döhner H, Döhner K, Guzman ML, Hassane DC, Roboz GJ, Grimwade D, Valk PJ, D'Andrea RJ, Carroll M, Park CY, Neuberg D, Levine R, Melnick AM, Mason CE. "Distinct evolution and dynamics of epigenetic and genetic heterogeneity in acute myeloid leukemia." Nature Medicine. 2016 Jul;22(7):792-9. http://dx.doi.org/10.1038/nm.4125 ; PubMed PMID: 27322744; PubMed Central PMCID: PMC4938719 , Jul-2016
Articles in Peer-reviewed Journals	Zook JM, Catoe D, McDaniel J, Vang L, Spies N, Sidow A, Weng Z, Liu Y, Mason CE, Alexander N, Henaff E, McIntyre AB, Chandramohan D, Chen F, Jaeger E, Moshrefi A, Pham K, Stedman W, Liang T, Saghbini M, Dzakula Z, Hastie A, Cao H, Deikus G, Schadt E, Sebra R, Bashir A, Truty RM, Chang CC, Gulbahce N, Zhao K, Ghosh S, Hyland F, Fu Y, Chaisson M, Xiao C, Trow J, Sherry ST, Zaranek AW, Ball M, Bobe J, Estep P, Church GM, Marks P, Kyriazopoulou-Panagiotopoulou S, Zheng GX, Schnall-Levin M, Ordonez HS, Mudivarti PA, Giorda K, Sheng Y, Rypdal KB, Salit M. "Extensive sequencing of seven human genomes to characterize benchmark reference materials." Nature Scientific Data. 2016 Jun 7;3:160025. http://dx.doi.org/10.1038/sdata.2016.25 ; PubMed PMID: 27271295; PubMed Central PMCID: PMC4896128 , Jun-2016